Mushfique Shahriar Aurnab

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Education

Daegu Gyeongbuk Institute of Science and Technology (DGIST) Daegu, Republic of Korea

B.S. student; intended focus in Computer Science and Electrical Engineering *Aug. 2024 – Present*

- Selected coursework: Probability and Mathematical Statistics with R, Artificial Intelligence Basics, Object-Oriented Programming, Data Structures, and System Programming.

Research Experience

Undergraduate Researcher, UGRP I DGIST

Comparative Study of Catastrophic Forgetting in CNNs and Vision Transformers *Spring 2026*

- Designed a nine-run sequential CIFAR-10 diagnostic across three class splits and three seeds, using linear probes at six depths to distinguish classifier-readout failure from deeper representation loss.
- At Task-B epoch 25, 79.7% of misclassified old-task examples remained linearly recoverable; refitting the full linear readout restored 74.6% mean old-task accuracy.
- Compared five replay-selection strategies and retained the negative result: random replay outperformed collision-, fate-, memorization-, and speed-based targeting in buffer-only recovery.

Selected Projects

GPX Motion Analysis | *Python, time-series alignment, Matplotlib* 2026

- Rebuilt a motion-data project as a tested CLI that aligns irregular GPX tracks on a shared grid while preserving segment boundaries and long gaps as missing evidence.
- Exports aligned samples, motion metrics, threshold sensitivity, summary JSON, and figures from deterministic synthetic fixtures; CI checks Python 3.11 and 3.14.

Vox Pop | *Python, MCP, FastEmbed* 2026

- Built a Python library, CLI, and MCP server for searching nine public discussion platforms; a semantic router selects relevant communities before retrieval.

Course Extractor | *Python, Requests, BeautifulSoup* 2026

- Implemented FOSE endpoint discovery with JSON-LD and HTML fallbacks; validated structured JSON and Markdown output on 508 Harvard DCE courses with zero reported page errors.

MapleStory Code Battle AI | *Python, OpenCV, minimax* 2025

- Converted screen captures into board state with template matching and HSV ownership detection, then selected moves with summed-area tables and 3-ply minimax with alpha-beta pruning.

Technical Skills

Languages: Python, C++
Machine learning and vision: PyTorch, torchvision, NumPy, OpenCV, Matplotlib, linear probing
Systems and tools: Git, pytest, Ruff, asynchronous HTTP APIs, MCP, CLI packaging, JSON and Markdown pipelines